

Problem Set 4

Diogn Lei R. Mortera

Contents

Problem Set 4	1
1. Prove or Disprove: $n^2 = \Omega(3n^2 + 2n + 5)$	2
2. Prove that $\lfloor \frac{n}{7} \rfloor = \Theta(n)$	3
3. Find the smallest nonnegative real r such that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^r)$ (and prove it)!	4
4. <i>Prove or Disprove:</i> $2^{n-2} = \Omega(2^n)$	6
5. <i>Prove or Disprove:</i> $2^{2n} = \Omega(2^n)$	6
6. <i>Prove or Disprove:</i> $2^{2^{n+1}} = \Omega(2^{2^n})$	6
7. Prove that $2^n n^{32} = \mathcal{O}(\frac{3^n}{n^{32}})$	6
8. Prove that $n \log(n \log(n \log n)) = \Theta(n \log n)$	7
9. Prove that $\log n! = \Theta(n \log n)$	8
10. Prove that $\log n! = \Theta(n \log n)$	9
10.1 If $f(n) = \Omega(g(n))$ and $g(n) = \Omega(h(n))$, then $f(n) = \Omega(h(n))$	9
10.2 If $f(n) = \mathcal{O}(g(n))$, then $f(n) + g(n) = \Theta(g(n))$	10
10.3 $f(n) = o(g(n))$ if and only if $\mathcal{O}(f(n)) \subsetneq \mathcal{O}(g(n))$	10
10.4 If $f(n) = \Theta(g(n))$, then the set $\Omega(f(n))$ is equal to the set $\Omega(g(n))$	14
10.5 At least one of the following is true: $f(n) = \mathcal{O}(g(n))$ or $f(n) = \Omega(g(n))$	14
10.6 $f(n) + g(n) = \Theta(\max(f(n), g(n)))$	14
10.7 $2^{f(n)+g(n)} = \Theta(2^{\max(f(n), g(n))})$	15
10.8 $\mathcal{O}(f(n)) \subsetneq \mathcal{O}(nf(n))$	15
10.9 $\mathcal{O}(f(n)) \subseteq \mathcal{O}(f(n)g(n))$	15
10.10 If $f(n) \leq g(n) + c$ for all positive integers n , then $f(n) = \mathcal{O}(g(n))$	15
10.11 If $\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} g(n) = \infty$ and $f(n)^a = \Theta(g(n)^b)$, then $\log f(n) = \Theta(\log g(n))$	16
10.12 If $f(n) = \Theta(n^3)$, then $f(n) \log f(n) = \Theta(n^3 \log n)$	17
10.13 $f(n)g(n) = \mathcal{O}(f(n)^2 + g(n)^2)$	20
10.14 $f(n)g(n) = \Omega(f(n)^2 + g(n)^2)$	21
11. <i>Prove or Disprove:</i> $\sum_{k=n}^{2n} \frac{1}{k} = \Theta(1)$	21

1. Prove or Disprove: $n^2 = \Omega(3n^2 + 2n + 5)$.

Proof.

This statement is true.

Let $c := \frac{1}{10}$ and $n_0 := 1$. Then:

$$0 \leq \frac{1}{10}(3n^2 + 2n + 5) \leq \frac{1}{10}(3n^2 + 2n^2 + 5n^2) \leq \frac{1}{10}(10n^2) = n^2$$

then by definition, $n^2 = \Omega(3n^2 + 2n + 5)$.

■

2. Prove that $\lfloor \frac{n}{7} \rfloor = \Theta(n)$

Proof.

Let $c_1 := 1, c_2 := \frac{1}{2}$ and $n_0 := 14$. Then by definition, $\lfloor \frac{n}{7} \rfloor = \Theta(n)$.

Recall the definition of $\lfloor \frac{n}{7} \rfloor$:

$$\left\lfloor \frac{n}{7} \right\rfloor \leq \frac{n}{7} < \left\lfloor \frac{n}{7} \right\rfloor + 1$$

$c_1 := 1$ works from the following inequality:

$$\left\lfloor \frac{n}{7} \right\rfloor \leq 1 \cdot \frac{n}{7}$$

$c_2 := \frac{1}{2}$ works because $\frac{n}{14} \geq 1$ at $n \geq 14$. Hence,

$$\frac{1}{2} \cdot \frac{n}{7} = \frac{n}{7} - \frac{1}{2} \cdot \frac{n}{7} \leq \frac{n}{7} - 1 \leq \left\lfloor \frac{n}{7} \right\rfloor$$

Then, by definition, $\lfloor \frac{n}{7} \rfloor = \Theta(n)$ (with constants c_1, c_2 , and n_0).

■

3. Find the smallest nonnegative real r such that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^r)$ (and prove it)!

We claim that $r = 33$.

Proof. We prove this by showing that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$ and $n^{32 + \sin(\frac{\pi n}{32})} \neq \mathcal{O}(n^r)$ for all $r < 33$.

We now prove the first statement. Choose $c := 1$ and $n_0 := 2$. Then for all $n \geq n_0$, using the fact

$$-1 \leq \sin \frac{\pi n}{32} \leq 1 \implies 31 \leq 32 + \sin \frac{\pi n}{32} \leq 33$$

We can get:

$$n^{32 + \sin \frac{\pi n}{32}} \leq n^{33} = 1 \cdot n^{33}$$

which shows that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$ (with constants c, n_0).

We will prove the second statement using contradiction. Suppose $n^{32 + \sin(\frac{\pi n}{32})} = \mathcal{O}(n^r)$. This assumes:

$$n^{32 + \sin(\frac{\pi n}{32})} \leq n^r$$

Note that $n^{32 + \sin(\frac{\pi n}{32})} = n^{33}$ for infinitely many values $n \geq n_0$. We therefore have another inequality for infinitely many n :

$$n^{32 + \sin(\frac{\pi n}{32})} = n^{33}$$

which contradicts with our first inequality.

■

We claim that $r = 33$.

Old Proof 3. We prove this by showing that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$ and $n^{32 + \sin(\frac{\pi n}{32})} \neq O(n^r)$ for all $r < 33$.

Choose $c := 1$ and $n_0 := 2$. Then for all $n \geq n_0$, using the fact

$$-1 \leq \sin \frac{\pi n}{32} \leq 1 \implies 31 \leq 32 + \sin \frac{\pi n}{32} \leq 33$$

We can get:

$$n^{32 + \sin \frac{\pi n}{32}} \leq n^{33} = 1 \cdot n^{33}$$

which shows that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$ (with constants c, n_0).

Now suppose $n^{32 + \sin(\frac{\pi n}{32})} = O(n^r)$. Then there would exist a c_1 and n_0 such that for all $n \geq n_0$:

$$n^{32 + \sin(\frac{\pi n}{32})} \leq n^{33}$$

But by definition, we also know:

$$n^r \leq n^{33}$$

But that last inequality implies that some nonconstant $n^{33-r} \leq c$ where $33 - r > 0$ which is a contradiction.

Therefore, $n^{32 + \sin(\frac{\pi n}{32})} \neq O(n^r)$.

■

We claim that $r = 33$.

Old Proof 2. We prove this by showing that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$ and $n^{32 + \sin(\frac{\pi n}{32})} = \omega(n^r)$ for all $r < 33$.

Choose $c := 1$ and $n_0 := 2$. Then for all $n \geq n_0$, using the fact

$$-1 \leq \sin \frac{\pi n}{32} \leq 1 \implies 31 \leq 32 + \sin \frac{\pi n}{32} \leq 33$$

We can get:

$$n^{32 + \sin \frac{\pi n}{32}} \leq n^{33} = 1 \cdot n^{33}$$

which shows that $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$ (with constants c, n_0).

Also, for $r < 33$, note that:

$$\lim_{n \rightarrow \infty} \frac{n^{32 + \sin \frac{\pi n}{32}}}{n^r} = \lim_{n \rightarrow \infty} n^{32 + \sin(\frac{\pi n}{32}) - r} = \infty$$

because $r < 33$ and $32 + \sin(\frac{\pi n}{32}) \leq 33$ *infinitely many times* as n grows due to periodicity of $\sin(\frac{\pi n}{32})$, we can say $32 + \sin(\frac{\pi n}{32}) - r > 0$ *infinitely many times* as n grows without bound. This implies $n^{32 + \sin(\frac{\pi n}{32})} = \omega(n^r)$ for all $r < 33$.

■

Claim. $r = 33$.

Proof. Let $c = 1$ and $n_0 = 2$. For all $n \geq n_0$, note that

$$\begin{aligned}\sin \frac{\pi n}{32} &\leq 1 \\ 32 + \sin \frac{\pi n}{32} &\leq 33\end{aligned}$$

Since $n \geq 1$, the function $x \mapsto n^x$ is non-decreasing in x , so

$$n^{32 + \sin \frac{\pi n}{32}} \leq n^{33} = 1 \cdot n^{33}.$$

Thus, by definition, $n^{32 + \sin \frac{\pi n}{32}} = \mathcal{O}(n^{33})$.

To see that $r = 33$ is smallest: since $\sin \frac{\pi n}{32} = 1$ infinitely often (e.g. whenever $\frac{n}{32}$ hits an odd multiple of $\frac{1}{2}$), the exponent $32 + \sin \frac{\pi n}{32}$ equals 33 for infinitely many n , so no $r < 33$ can satisfy $n^{32 + \sin \frac{\pi n}{32}} \leq c \cdot n^r$ for all sufficiently large n . Hence $r = 33$ is the smallest such value. ■

4. *Prove or Disprove:* $2^{n-2} = \Omega(2^n)$

5. *Prove or Disprove:* $2^{2n} = \Omega(2^n)$

6. *Prove or Disprove:* $2^{2^{n+1}} = \Omega(2^{2^n})$

7. Prove that $2^n n^{32} = \mathcal{O}\left(\frac{3^n}{n^{32}}\right)$

8. Prove that $n \log(n \log(n \log n)) = \Theta(n \log n)$

Proof:

We want to prove that there exists some c_1, c_2 , and n_0 such that for all $n \geq n_0$:

$$c_1 n \log n \leq n \log(n \log(n \log n)) \leq c_2 n \log n$$

We use two facts:

- $\log \log n \leq \log n$ for $n \geq n_1$.
- $2 \log n \leq n$ for $n \geq n_2$ (since $\log n = o(n)$)

Let $c_1 = 1$, $c_2 = 2$, and $n_0 = \max(n_1, n_2, 2)$. For all $n \geq n_0$, note that

$$0 < 1 \cdot n \log n \leq n \log(n \log(n \log n)) \leq 2 \cdot n \log n$$

The choice of c_1 comes from:

$$1 \cdot n \log n = n \log n \leq n \log(n \log(n \log n))$$

Because $n \leq n \log(n \log n)$.

Meanwhile, the choice of c_2 comes from:

$$\begin{aligned} n \log(n \log(n \log n)) &= n[\log(n) + \log(\log(n \log n))] \\ &= n[\log(n) + \lceil \log(\log(n) + \log(\log n)) \rceil] \\ &\leq n[\log(n) + \lceil \log(\log(n) + \log n) \rceil] \\ &= n[\log(n) + \log(n)] \\ &= 2 \cdot n \log(n) \end{aligned}$$

Hence, by definition, $n \log(n \log(n \log n)) = \Theta(n \log n)$. ■

9. Prove that $\log n! = \Theta(n \log n)$

Proof:

Let $c_1 = \frac{1}{8}$, $c_2 = 1$, and $n_0 = 4$. For all $n \geq n_0$, we can show that

$$0 < \frac{1}{8} \cdot n \log n \leq \log n! \leq 1 \cdot n \log n$$

which allows us to conclude $\log n! = \Theta(n \log n)$ (with constants c_1, c_2 , and n_0).

We can show that:

$$\left(\frac{n}{2} - 1\right) \log \frac{n}{2} \leq \left\lfloor \frac{n}{2} \right\rfloor \log \left\lceil \frac{n}{2} \right\rceil \leq \log \left(\prod_{i=\lceil \frac{n}{2} \rceil}^n \left\lceil \frac{n}{2} \right\rceil \right) \leq \log \left(\prod_{i=\lceil \frac{n}{2} \rceil}^n i \right) \leq \log n!$$

Note that for $n \geq 4$:

$$\begin{aligned} \frac{n}{4} &\leq \frac{n}{2} - 1 \\ \frac{1}{2} \log n &\leq \log \frac{n}{2} \end{aligned}$$

Hence:

$$\frac{1}{8} n \log n \leq \left(\frac{n}{2} - 1\right) \log \frac{n}{2}$$

Meanwhile $c_2 = 1$ comes from:

$$\log n! = \sum_{i=1}^n \log i \leq \sum_{i=1}^n \log n = n \log n = 1 \cdot n \log n$$

■

In the following, f , g , and h are positive functions taking in positive integer arguments and a , b and c are positive constants.

Formally *prove or disprove* each of the following:

10. Prove that $\log n! = \Theta(n \log n)$

Proof.

10.1 If $f(n) = \Omega(g(n))$ and $g(n) = \Omega(h(n))$, then $f(n) = \Omega(h(n))$.

Proof.

By definition of Ω , we know there exist constants c_1 and n_1 such that for all $n \geq n_1$:

$$0 \leq c_1 g(n) \leq f(n)$$

Similarly, we know there exist constants c_2 and n_2 such that for all $n \geq n_2$:

$$0 \leq c_2 h(n) \leq g(n)$$

Define $n_0 := \max(n_1, n_2)$. Then for all n , $n \geq n_0$, after multiplying the inequalities:

$$0 < c_1 g(n) \leq c_1 (c_2 h(n)) = (c_1 c_2) h(n) \leq f(n)$$

Therefore, by definition, $f(n) = \Omega(h(n))$ (with constants $c_1 c_2, n_0$).

■

10.2 If $f(n) = \mathcal{O}(g(n))$, then $f(n) + g(n) = \Theta(g(n))$.

Proof:

By definition of $\mathcal{O}$, since $f(n) = \mathcal{O}(g(n))$, there exists c_3 and n_1 such that for all $n \geq n_1$:

$$0 < f(n) \leq c_3 g(n)$$

By adding $g(n)$ to all parts of the inequality, we get:

$$1 \cdot g(n) < f(n) + g(n) \leq c_3 g(n) + g(n) = (c_3 + 1)g(n)$$

which allows us to conclude that $f(n) + g(n) = \Theta(g(n))$ (with constants $1, c_3 + 1, n_1$)

■

Old Proof:

We want to show that there exists c_1, c_2 and n_0 such that for all $n \geq n_0$:

$$0 < c_1 g(n) \leq f(n) \leq c_2 g(n)$$

By definition of $\mathcal{O}$, since $f(n) = \mathcal{O}(g(n))$, there exists c_3 and n_1 such that for all $n \geq n_1$:

$$0 < f(n) \leq c_3 g(n)$$

Let $c_1 := 1$, $c_2 := c_3 + 1$, and $n_0 := n_1$.

By adding $g(n)$ to all parts of the inequality, we get:

$$1 \cdot g(n) < f(n) + g(n) \leq c_3 g(n) + g(n) = (c_3 + 1)g(n)$$

as desired.

■

10.3 $f(n) = o(g(n))$ if and only if $\mathcal{O}(f(n)) \subsetneq \mathcal{O}(g(n))$.

Proof. This statement is false.

Let

$$f(n) = \begin{cases} n & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$$
$$g(n) = n$$

We want to prove that $\mathcal{O}(f(n)) \subseteq \mathcal{O}(g(n))$, $\mathcal{O}(f(n)) \neq \mathcal{O}(g(n))$, but $f(n) \neq o(g(n))$.

$$\mathcal{O}(f(n)) \subseteq \mathcal{O}(g(n))$$

Let $h \in \mathcal{O}(f(n))$. Then there exists $c := 1, n_0$ such that for all $n \geq n_0$:

$$h(n) \leq n = 1 \cdot g(n)$$

Since $h(n)$ was chosen arbitrarily, then $\mathcal{O}(f(n)) \subseteq \mathcal{O}(g(n))$.

$$\mathcal{O}(f(n)) \neq \mathcal{O}(g(n))$$

Suppose $\mathcal{O}(f(n)) = \mathcal{O}(g(n))$. That is, since $\mathcal{O}(g(n)) \subseteq \mathcal{O}(f(n))$. Let $h \in \mathcal{O}(f(n))$. Note that for odd n :

$$h(n) \leq cg(n) = c \cdot 1 = c$$

But $h(n)$ is an unbounded variable, hence this is a contradiction. Since h was arbitrary, $\mathcal{O}(f(n)) \subsetneq \mathcal{O}(g(n))$ implying $\mathcal{O}(f(n)) \neq \mathcal{O}(g(n))$.

$$f(n) \neq o(g(n))$$

That is,

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{n}{n} = 1 \neq 0$$

for even n . ■

Old Proof 2.

This statement is true.

($\implies$)

Assume f and g are functions such that $f(n) = o(g(n))$. We want to show that $O(f(n)) \subseteq O(g(n))$ and $O(f(n)) \neq O(g(n))$ by contradiction.

Suppose that $O(f(n)) = O(g(n))$. Since trivially, $g(n) = O(g(n))$, it follows that $g(n) = O(f(n))$ due to equality. By definition, this means there is some $c > 0$ and n_0 such that for all $n \geq n_0$:

$$g(n) \leq cf(n)$$

$$\frac{f(n)}{g(n)} \geq \frac{1}{c}$$

It follows from here that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \geq \frac{1}{c} > 0$, implying that $f(n) \neq o(g(n))$. We have reached a contradiction.

($\impliedby$)

Assume f and g are functions such that $O(f(n)) \subseteq O(g(n))$ and $O(f(n)) \neq O(g(n))$. We want to show that $f(n) = o(g(n))$ by contradiction.

Suppose $f(n) \neq o(g(n))$. Then there exists a constant c such that for all $n \geq n_1$:

$$\frac{f(n)}{g(n)} \geq c \iff f(n) \geq c \cdot g(n) \iff g(n) \leq \frac{1}{c}f(n)$$

implying $g(n) = O(f(n))$ (for constants $\frac{1}{c}, n_1$). Therefore, $O(g(n)) \subseteq O(f(n))$. Together with Therefore, $O(f(n)) \subseteq O(g(n))$, this implies $O(f(n)) = O(g(n))$ which is a contradiction.

■

Old Proof.

($\implies$)

We know that for all c , there exists an n_0 such that for all $n \geq n_0$:

$$f(n) < cg(n)$$

We will prove $\mathcal{O}(f(n)) \subseteq \mathcal{O}(g(n))$ and $\mathcal{O}(f(n)) \neq \mathcal{O}(g(n))$.

Let $f_1 \in \mathcal{O}(f(n))$. From the fact above, it follows that $f_2 \in \mathcal{O}(f(n))$. Hence, $\mathcal{O}(f(n)) \subseteq \mathcal{O}(g(n))$.

Now, let $g_1 = \frac{1}{c}g(n)$ be some function in $\mathcal{O}(g(n))$. Then:

$$g_1(n) = \frac{1}{c} \cdot c \cdot g(n) = 1 \cdot g(n) \not\leq g(n)$$

Hence, we have found an element in $\mathcal{O}(g(n))$ not in $\mathcal{O}(f(n))$.

($\Leftarrow$)

We know that $\mathcal{O}(f(n)) \subsetneq \mathcal{O}(g(n))$. We want to show $f(n) = o(g(n))$ by contradiction.

Suppose that $f_2(n) \geq cg(n)$.

Rearranging:

$$\frac{1}{c}g(n) \leq f_2(n)$$

implying $f_2(n) \in \mathcal{O}(g(n))$.

Since $f_2(n)$ was arbitrary, we have $\mathcal{O}(f(n)) \subseteq \mathcal{O}(g(n))$.

This is a contradiction, which means $c \cdot g(n) < f(n)$ or $g(n) = o(f(n))$ as desired.

■

10.4 If $f(n) = \Theta(g(n))$, then the set $\Omega(f(n))$ is equal to the set $\Omega(g(n))$.

10.5 At least one of the following is true: $f(n) = \mathcal{O}(g(n))$ or $f(n) = \Omega(g(n))$.

10.6 $f(n) + g(n) = \Theta(\max(f(n), g(n)))$.

Proof. Let $c_1 := 1$ and $c_2 := 2$. We will show that there exists some n_0 such that $n \geq n_0$.

$$c_1 \max(f(n), g(n)) \leq f(n) + g(n) \leq c_2 \max(f(n), g(n))$$

to conclude $\Theta(\max(f(n), g(n)))$ (with constants c_1, c_2, n_0).

Note that $f(n) \leq \max(f(n), g(n))$ and $g(n) \leq \max(f(n), g(n))$

Hence:

$$f(n) + g(n) \leq 2 \max(f(n), g(n)) = c_2 \cdot \max(f(n), g(n))$$

Also, note that

$$c_1 \cdot \max(f(n), g(n)) = 1 \cdot \max(f(n), g(n)) \leq f(n) + g(n)$$

■.

Old Proof.

We want to show that there exists a c_1, c_2 , and n_0 such that for all $n \geq n_0$:

$$c_1 \max(f(n), g(n)) \leq f(n) + g(n) \leq c_2 \max(f(n), g(n))$$

Setting $c_1 = 1$ and $c_2 = 2$, we have:

$$1 \cdot \max(f(n), g(n)) \leq f(n) + g(n) \leq 2 \cdot \max(f(n), g(n))$$

The value of c_1 is trivial.

The value of c_2 is derived from definition of:

$$f(n) \leq \max(f(n), g(n))$$

$$g(n) \leq \max(f(n), g(n))$$

$$f(n) + g(n) \leq 2 \cdot \max(f(n), g(n))$$

■.

$$10.7 \quad 2^{f(n)+g(n)} = \Theta(2^{\max(f(n),g(n))}).$$

Proof.

False. Let $f(n) = n$ and $g(n) = 2n$.

Then

$$2^{f(n)+g(n)} = 2^{3n} = 8^n$$

$$2^{\max(f(n),g(n))} = 2^{2n} = 4^n$$

Clearly, $8^n \neq \Theta(4^n)$.

■.

$$10.8 \quad \mathcal{O}(f(n)) \subsetneq \mathcal{O}(nf(n)).$$

Proof.

We want to prove that for some g , if $g(n) = \mathcal{O}(f(n))$, then it is also in $\mathcal{O}(nf(n))$.

By definition, there exists some c and n_0 such that

$$g(n) \leq cf(n) \leq cnf(n)$$

.

Thus, $g(n) = \mathcal{O}(nf(n))$.

Now, take an example $h(n) = nf(n)$. Then clearly $nf(n) = \mathcal{O}(nf(n))$ but $nf(n) \notin \mathcal{O}(f(n))$.

Therefore, $\mathcal{O}(f(n)) \neq \mathcal{O}(nf(n))$.

■

$$10.9 \quad \mathcal{O}(f(n)) \subseteq \mathcal{O}(f(n)g(n)).$$

Proof.

False. Let $f(n) = n$ and $g(n) = \frac{1}{n}$.

Then some $h(n) = n$ belongs in $\mathcal{O}(n)$ but not in $\mathcal{O}(f(n)g(n)) = \mathcal{O}(1)$. We have found a contradiction.

■

$$10.10 \quad \text{If } f(n) \leq g(n) + c \text{ for all positive integers } n, \text{ then } f(n) = \mathcal{O}(g(n)).$$

Proof. False. $f(n) := 2$, Let $g(n) := \frac{1}{n}$, $c := 1$, $n_0 := 1$.

Clearly, $2 \leq \frac{1}{n} + 1$ for all $n \geq n_0$, but $2 \neq \mathcal{O}(\frac{1}{n})$.

■

Old Proof 2.

Let $c := 2$ and $n_0 := 1$. We can see that:

$$f(n) \leq g(n) + c \leq g(n) + g(n) = c_2 g(n)$$

for all $n \geq n_0$, which would imply that $f(n) = \mathcal{O}(g(n))$ (with constants c, n_0).

Old Proof.

We want to show that there exists a c_1, c_2 , and n_0 such that for all $n \geq n_0$:

$$c_1 g(n) \leq f(n) \leq c_2 g(n)$$

Let $c_1 = 1$, and c_2 be a positive integer. We choose an n_0 such that it satisfies $g(n_0) \geq c_2$.
Thus:

$$1 \cdot g(n) \leq g(n) + c \leq g(n) + g(n) \leq 2 \cdot g(n)$$

as desired.

■

10.11 If $\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} g(n) = \infty$ and $f(n)^a = \Theta(g(n)^b)$, then $\log f(n) = \Theta(\log g(n))$.

Proof.

NOT DONE

NOT DONE

NOT DONE

10.12 If $f(n) = \Theta(n^3)$, then $f(n) \log f(n) = \Theta(n^3 \log n)$.

Proof.

Let f be some function in $\Theta(n^3)$. Then there exists c_1, c_2, n_0 such that for all $n \geq n_0$:

$$c_1 n^3 \leq f(n) \leq c_2 n^3$$

Choose $c_3 := 3c_1, c_4 := 6c_2$, and $n_0 := (c_2)^{\frac{1}{3}}$. We'll show that

$$c_3 n^3 \log n \leq f(n) \log f(n) \leq c_4 n^3 \log n$$

for all $n \geq n_0$, which will imply that $f(n) \log f(n) = \Theta(n^3 \log n)$.

Note that taking the log of both sides of $f(n) \leq c_2 n^3$:

$$\begin{aligned} \log f(n) &\leq \log(c_2 n^3) = \log(c_2) + \log(n^3) \\ &\leq 2 \log(n^3) = 6 \log(n) \end{aligned}$$

Multiplying both sides by $f(n)$:

$$\begin{aligned} f(n) \log f(n) &\leq f(n) \cdot 6 \log(n) \\ &\leq (6c_2) n^3 \log(n) \end{aligned}$$

Meanwhile:

$$\begin{aligned} c_1 n^3 &\leq f(n) \\ \log(c_1) + \log(n^3) &\leq \log f(n) \\ 3 \log n = \log(n^3) &\leq \end{aligned}$$

Multiplying both sides by $f(n)$:

$$(3c_1) n^3 \log n = f(n) 3 \log n \leq f(n) \log f(n)$$

■

Old Proof 2.

Let f be some function in $\Theta(n^3)$. Then there exists c_1, c_2, n_0 such that for all $n \geq n_0$:

$$c_1 n^3 \leq f(n) \leq c_2 n^3$$

Choose $c_3 := 3c_1, c_4 := 12c_2$, and $n_0 := (c_2)^{\frac{1}{3}}$. We'll show that

$$c_3 n^3 \log n \leq f(n) \log f(n) \leq c_4 n^3 \log n$$

for all $n \geq n_0$, which will imply that $f(n) \log f(n) = \Theta(n^3 \log n)$.

Taking the log of our given inequality, we see that:

$$\begin{aligned} \log(c_1 n^3) &\leq \log f(n) \leq \log(c_2 n^3) \\ (\Leftrightarrow) \log c_1 + \log n^3 &\leq \log f(n) \leq \log c_2 + \log n^3 \\ (\Leftrightarrow) \log n^3 \leq \log c_1 + \log n^3 &\leq \log f(n) \leq \log c_2 + \log n^3 \leq 2 \log n^3 \end{aligned}$$

Implying:

$$\begin{aligned} 3 \log n = \log n^3 &\leq \log f(n) \leq 6 \log n \\ 3f(n) \log n &\leq f(n) \log f(n) \leq 6f(n) \log n \\ (3c_1 n^3) \log n &\leq f(n) \log f(n) \leq 6 \cdot (2c_2 n^3) \log n = (12c_2) n^3 \log n \end{aligned}$$

.

■

Old Proof.

We want to prove that there exists some c_3, c_4 and n_0 such that for all $n \geq n_0$:

$$c_3 n^3 \log n \leq f(n) \log f(n) \leq c_4 n^3 \log n$$

Let f be some function in $\Theta(n^3)$. Then there exists c_1, c_2, n_0 such that for all $n \geq n_0$:

$$c_1 n^3 \leq f(n) \leq c_2 n^3$$

We can let $c_3 := 3c_1, c_4 := 12c_2$, and $n_0 := (c_2)^{\frac{1}{3}}$ and by definition, $f(n) \log f(n) = \Theta(n^3 \log n)$.

These constants are obtained through the following:

$$\begin{aligned} \log(c_1 n^3) &\leq \log f(n) \leq \log(c_2 n^3) \\ \log c_1 + \log n^3 &\leq \log f(n) \leq \log c_2 + \log n^3 \\ \log n^3 &\leq \log c_1 + \log n^3 \leq \log f(n) \leq \log c_2 + \log n^3 \leq 2 \log n^3 \end{aligned}$$

For clarity:

$$\begin{aligned} \log n^3 &\leq \log f(n) \leq 2 \log n^3 \\ 3 \log n &\leq \log f(n) \leq 6 \log n \\ 3f(n) \log n &\leq f(n) \log f(n) \leq 6f(n) \log n \\ (3c_1 n^3) \log n &\leq f(n) \log f(n) \leq 6 \cdot (2c_2 n^3) \log n = 12c_2 n^3 \log n \end{aligned}$$

■

$$10.13 \quad f(n)g(n) = \mathcal{O}(f(n)^2 + g(n)^2).$$

Proof.

We want to prove by definition of $\mathcal{O}$.

Let $c := \frac{1}{2}$ and $n_0 := 1$.

Then:

$$f(n)g(n) \leq \frac{1}{2}(f(n)^2 + g(n)^2)$$

$$2f(n)g(n) \leq f(n)^2 + g(n)^2$$

$$f(n)^2 + g(n)^2 \geq 2f(n)g(n)$$

$$f(n)^2 - 2f(n)g(n) + g(n)^2 \geq 0$$

$$(f(n) - g(n))^2 \geq 0$$

which is true by the trivial inequality.

Hence,

$$f(n)g(n) \leq \frac{1}{2}f(n)g(n) = c \cdot f(n)g(n)$$

.

Therefore, by definition of $\mathcal{O}$, $f(n)g(n) = \mathcal{O}(f(n)^2 + g(n)^2)$.

■

Old Proof 1.

We want to prove by definition of $\mathcal{O}$.

Let $c := \frac{1}{2}$ and $n_0 := 1$. Then

$$f(n)g(n) = \mathcal{O}(f(n)^2 + g(n)^2).$$

This constant was obtained using the trivial inequality:

$$(f(n) - g(n))^2 \geq 0$$

$$f(n)^2 + g(n)^2 - 2f(n)g(n) \geq 0$$

$$f(n)^2 + g(n)^2 \geq 2f(n)g(n)$$

$$\frac{1}{2}(f(n)^2 + g(n)^2) \geq f(n)g(n)$$

$$f(n)g(n) \leq \frac{1}{2}(f(n)^2 + g(n)^2)$$

■

10.14 $f(n)g(n) = \Omega(f(n)^2 + g(n)^2)$.

Proof.

False. Let $f(n) = n$ and $g(n) = \frac{1}{n}$. Then $f(n)g(n) = 1$ and $f(n)^2 + g(n)^2 = n^2 + \frac{1}{n^2}$

We want to show this is not in $\Omega(f(n)^2 + g(n)^2)$. Which is true because no constant c will satisfy:

$$c \left(n^2 + \frac{1}{n^2} \right) \leq 1$$

because $n^2 + \frac{1}{n^2} \rightarrow \infty$ as $n \rightarrow \infty$.

■

11. *Prove or Disprove:* $\sum_{k=n}^{2n} \frac{1}{k} = \Theta(1)$

11. *Prove or Disprove:* $\sum_{k=n}^{2n} \frac{1}{k} = \Theta(1)$

Proof.

Choose $c_1 := \frac{1}{2}$, $c_2 := 2$, and $n_0 := 1$. We'll show that

$$c_1 \leq \sum_{k=n}^{2n} \frac{1}{k} \leq c_2$$

for all $n \geq n_0$, which will imply that $\sum_{k=n}^{2n} \frac{1}{k} = \Theta(1)$.

Note that $\frac{1}{2n+2} \leq \frac{1}{2n}$, so

$$\frac{1}{2} = \frac{n+1}{2n+2} = \sum_{k=n}^{2n} \frac{1}{2n+2} \leq \sum_{k=n}^{2n} \frac{1}{k}$$

Note that $\frac{1}{n} \leq \frac{2}{n+1}$, so

$$\sum_{k=n}^{2n} \frac{1}{k} \leq \sum_{k=n}^{2n} \frac{2}{n+1} \leq \frac{2(n+1)}{n+1} = 2$$

■

Proof.

We want to show there exists c_1, c_2 , and n_0 such that for all $n \geq n_0$:

$$c_1 \leq \sum_{k=n}^{2n} \frac{1}{k} \leq c_2$$

Choose $c_1 := \frac{1}{2}$, $c_2 := 2$, and $n_0 := 1$. Then by definition, $\sum_{k=n}^{2n} \frac{1}{k} = \Theta(1)$.

We obtained these constants from the following inequalities:

$$\begin{aligned} \frac{1}{2n+2} &\leq \frac{1}{2n} \\ \frac{1}{n} &\leq \frac{2}{n+1} \quad (n+1 \leq 2n \text{ at } n := n_0 \geq 1) \end{aligned}$$

Hence:

$$\begin{aligned} \frac{1}{2} = \frac{n+1}{2n+2} &= \sum_{k=n}^{2n} \frac{1}{2n+2} \leq \sum_{k=n}^{2n} \frac{1}{k} \\ \sum_{k=n}^{2n} \frac{1}{k} &\leq \sum_{k=n}^{2n} \frac{2}{n+1} \leq \frac{2(n+1)}{n+1} = 2 \end{aligned}$$

■.